Course Registration System

Final Report: Project Plan, Use Case, BPM, ERD, Screen Layouts and Database Tables

Auralis Team

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1 Project Plan

1.1 Problem and Objectives

Course registration puts heavy pressure on university staff at the start of every semester. When it is handled with paper forms or separate spreadsheets, each request is checked by hand (prerequisites, free seats, timetable conflicts), so errors and delays are common and managers cannot see current enrollment.

- Analyze the users, main processes and business rules of course registration.
- Model the processes and requirements with BPM and Use Case diagrams.
- Design the data with Chen and IE ERDs and database tables.
- Show the main screens and keep all models consistent.

1.2 Scope and Users

Scope area	What it includes
User accounts and login	Role-based access for students, staff, managers and administrators
Course and section management	Courses, sections, instructors, rooms, times and seat capacity
Registration and reports	Register, drop and waitlist with automatic checks; enrollment reports

Out of scope. Tuition payment, grades and transcripts, and software implementation. This project covers system analysis and design only.

User	Needs from the system
Student	Searches courses, registers and drops, checks status
Staff	Registrar and instructors: manage sections and class lists
Manager	Department head: monitors seats and enrollment
System Administrator	Manages user accounts and access

1.3 Business Rules

ID	Rule
BR-1	A student can register only if the prerequisite course is passed.
BR-2	A section cannot exceed its capacity; a full section offers the waitlist.
BR-3	A student cannot register for two classes at the same time.
BR-4	A course can be dropped only before the drop deadline.

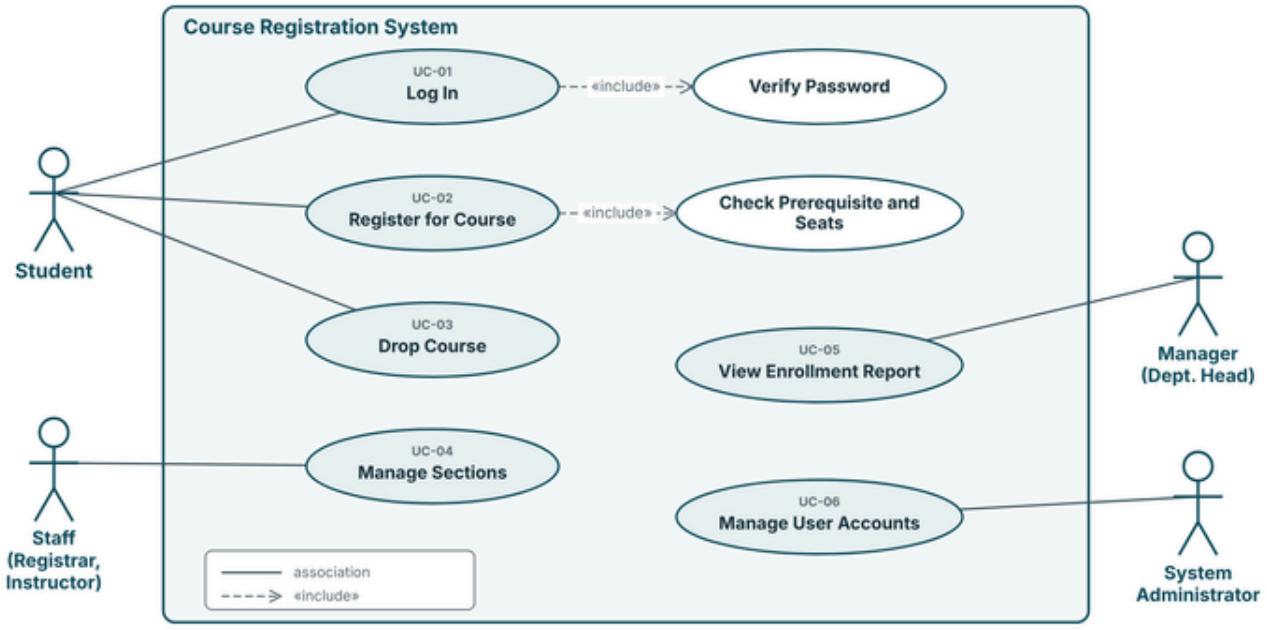
1.4 Timeline

Week	Dates	Activity
Week 1	5 - 11 Oct	Project plan and requirements
Week 2	12 - 18 Oct	Business process models
Week 3	19 - 25 Oct	Use case diagram and specifications
Week 4	26 Oct - 1 Nov	ERDs, database tables and review

Explanation The plan explains why the system is needed, what it covers, who uses it and who does the work. The business rules are enforced in the use cases, BPMs and database tables that follow.

2 Use Case Modeling

2.1 Use Case Diagram



Explanation Four actors use six use cases. The Student logs in (including password verification), registers for a course (including a check of the prerequisite and seats) and drops a course. Staff manage sections, the Manager views enrollment reports and the System Administrator manages user accounts.

2.2 Use Case Specifications

UC-02 Register for Course

Actor	Student
Precondition	The student is logged in and the registration period is open.
Main flow	1. The student searches and selects a section. 2. The system checks the prerequisite. 3. The system checks that a seat is free. 4. The system checks the time conflict. 5. The system creates the registration, adds 1 to seats_taken and shows a confirmation.
Alternative flow	Prerequisite missing or time conflict: the request is rejected with the reason. Section full: the system offers the waitlist.
Postcondition	A REGISTRATION record exists with status Registered.

UC-03 Drop Course

Actor	Student
Precondition	The student has a registration with status Registered or Waitlisted.
Main flow	1. The student selects the course and confirms the drop. 2. The system checks the drop deadline. 3. The system sets the status to Dropped and releases the seat. 4. The system promotes the first waitlisted student, if any. 5. The system shows a confirmation.
Alternative flow	Deadline passed: the drop is refused and the student is told why.
Postcondition	The registration status is Dropped and the seat is free again.

UC-04 Manage Sections

Actor	Staff (Registrar, Instructor)
Precondition	The staff member is logged in.
Main flow	1. Staff open the Manage Sections screen. 2. Staff create or edit a section: course, day and time, room, capacity and instructor. 3. The system checks room and instructor conflicts. 4. The system saves the section.
Alternative flow	Conflict found: the system shows a message and staff change the time or room.
Postcondition	The SECTION record is saved and students can register for it.

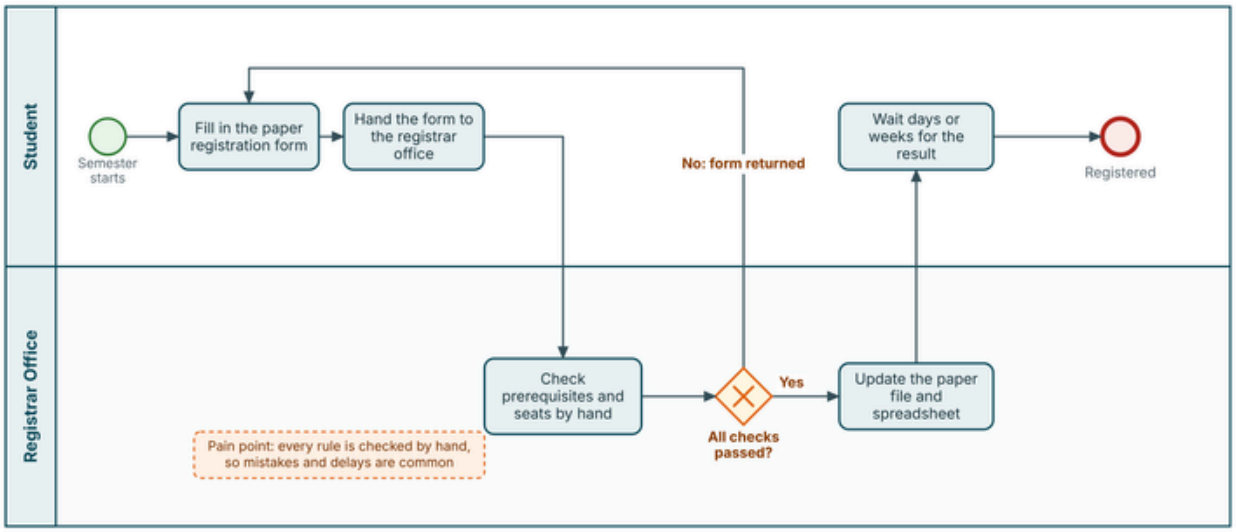
UC-05 View Enrollment Report

Actor	Manager (Department Head)
Precondition	The manager is logged in.
Main flow	1. The manager opens the report page and selects a semester. 2. The system reads capacity and seats_taken of every section. 3. The system shows seats used, full sections and waitlisted students.
Alternative flow	No data for the semester: the system shows an empty report message.
Postcondition	The report is displayed; no data is changed.

Explanation Each specification lists the actor, the conditions before and after, the main step-by-step flow and what happens when a check fails. UC-02 and UC-03 follow the business rules BR-1 to BR-4 and UC-04 and UC-05 serve staff and managers.

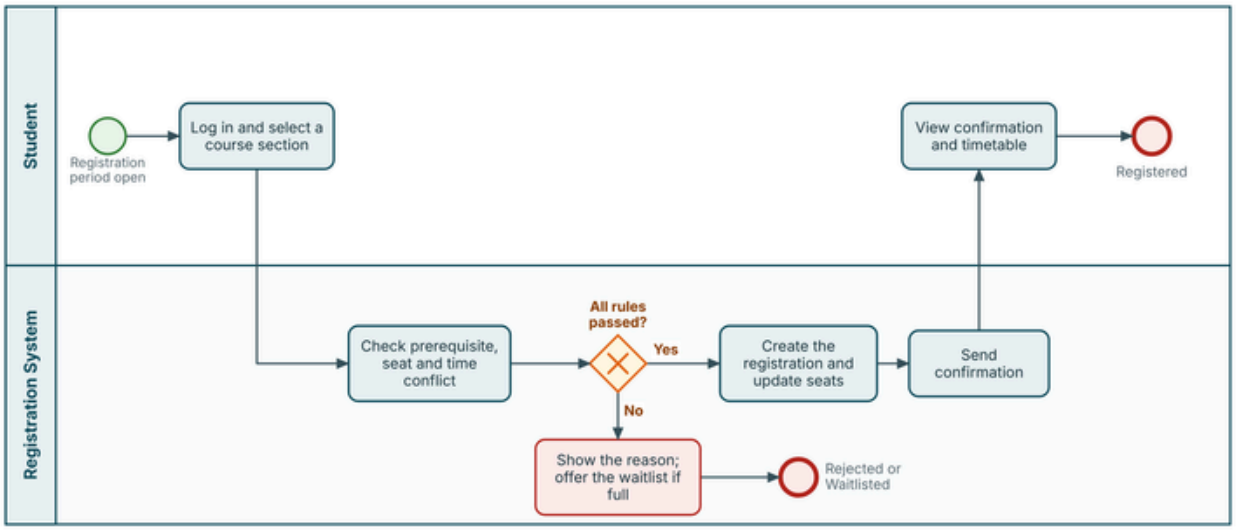
3 Business Process Models

3.1 BPM 1: Current Registration Process (As-Is)



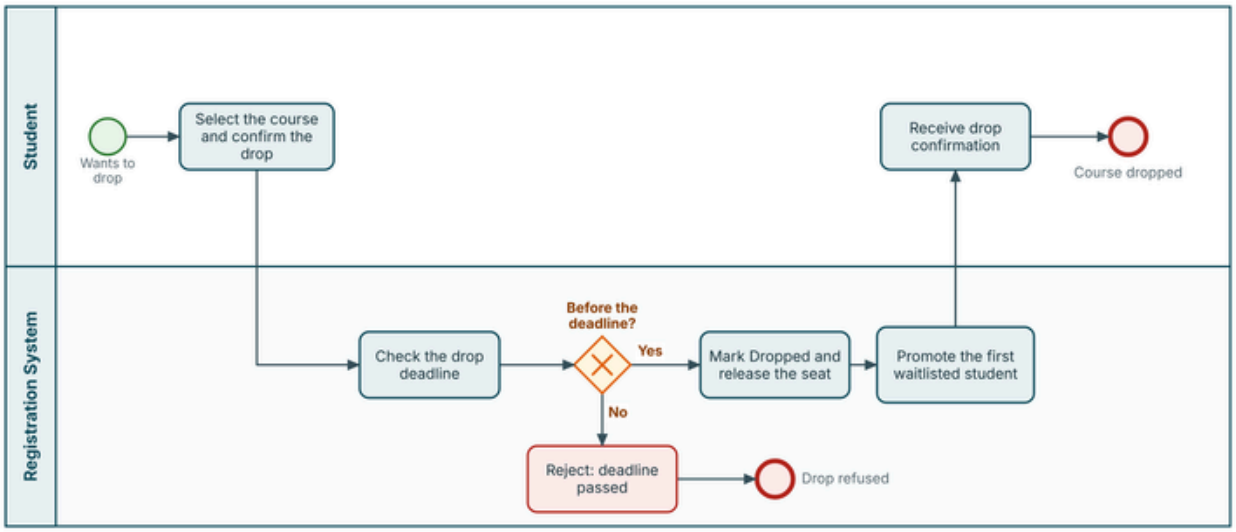
Explanation Today the student fills in a paper form and hands it to the registrar office. Staff check prerequisites and seats by hand, return the form if a check fails and update paper files. The student waits days or weeks for the result.

3.2 BPM 2: Register for a Course (To-Be)



Explanation In the proposed system the student selects a section and the system checks the prerequisite, seat and time conflict at once. If all rules pass it creates the registration, updates the seats and confirms; otherwise it shows the reason or offers the waitlist.

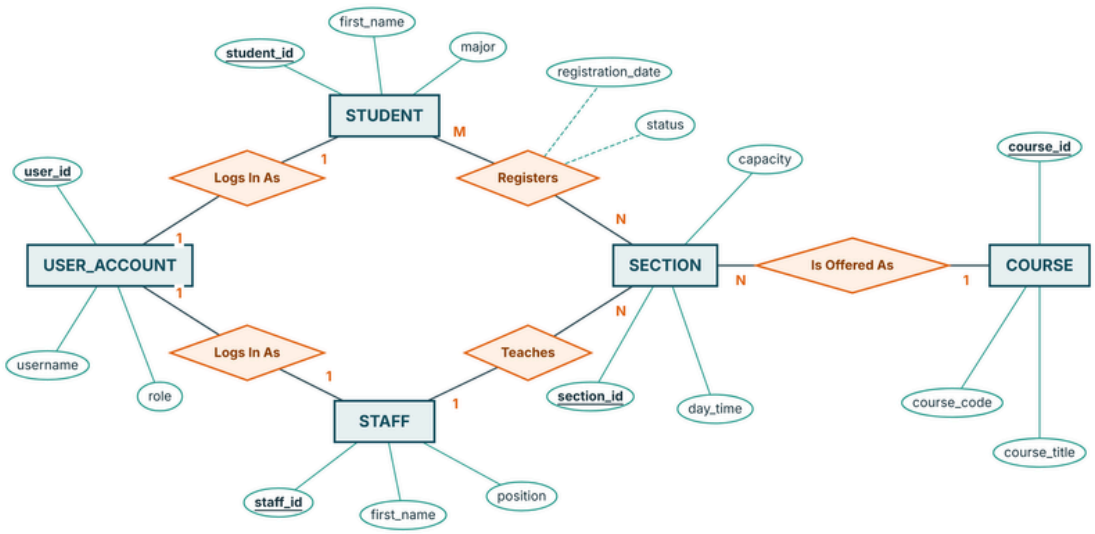
3.3 BPM 3: Drop Course



Explanation The system checks the drop deadline. Before the deadline it marks the registration Dropped, releases the seat and promotes the first waitlisted student; after the deadline the drop is refused. The student then receives a confirmation.

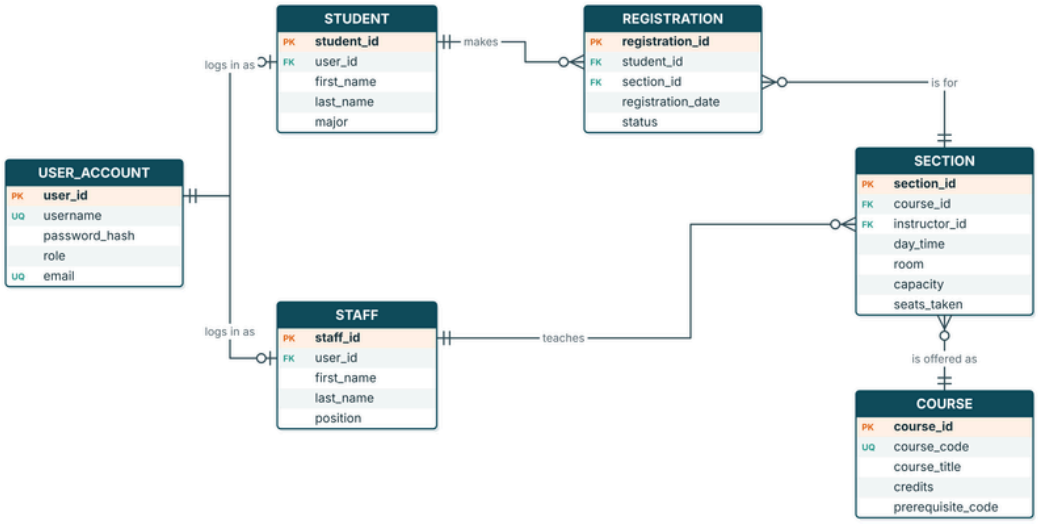
4 Entity Relationship Diagrams

4.1 Chen ERD



Explanation A USER_ACCOUNT logs in as one STUDENT or one STAFF member (1:1). A STUDENT registers for many SECTIONs and a SECTION has many students (M:N); the relationship carries registration_date and status. A STAFF member teaches many SECTIONs, and a COURSE is offered as many SECTIONs (1:N). Underlined attributes are primary keys; only the main attributes are shown.

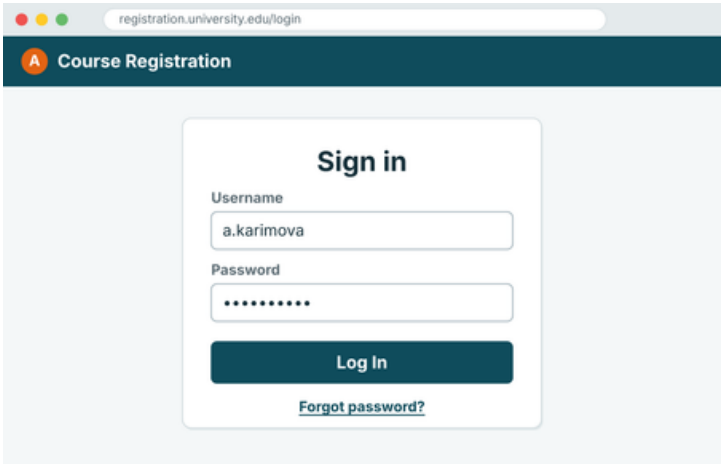
4.2 IE (Crow's Foot) ERD



Explanation The IE ERD turns the Chen model into six tables. REGISTRATION resolves the many-to-many link between STUDENT and SECTION, and SECTION holds foreign keys to COURSE and to STAFF (the instructor). PK = primary key, FK = foreign key, UQ = unique; the crow's foot marks the "many" side.

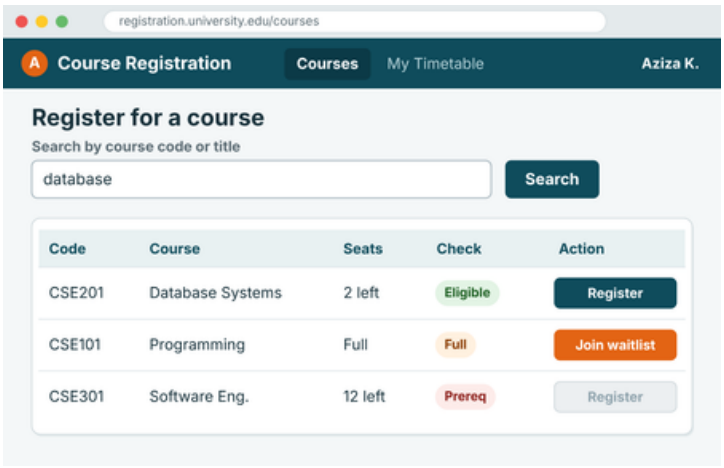
5 Screen Layouts

5.1 SCR-1 Login (UC-01)



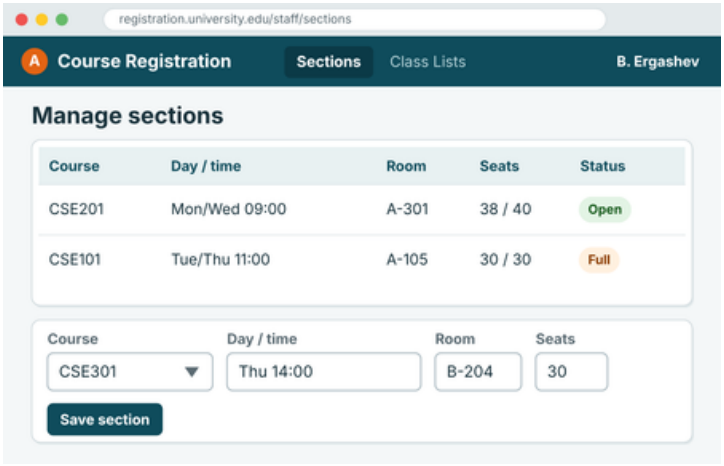
Explanation The login page is used by every role. After a successful login the system opens the page that matches the user's role; "Forgot password?" starts a password reset.

5.2 SCR-2 Register for a Course (UC-02)



Explanation Each section shows the seats left and the result of the automatic check (Eligible, Full or Prerequisite missing). The action button changes to Register, Join waitlist or a disabled button.

5.3 SCR-3 Manage Sections (UC-04)



Explanation Staff see the sections with time, room and seats, and can add a new section with its course, day and time, room and capacity. The system checks conflicts when it is saved.

6 Database Tables

6.1 Database Tables 1: USER_ACCOUNT, STUDENT and STAFF

USER_ACCOUNT — Login account of every user

PK user_id INT	UQ username VARCHAR(30)	password_hash VARCHAR(255)	role VARCHAR(15)	UQ email VARCHAR(80)
1	a.karimova	$2b$12$K9x...	Student	a.karimova@uni.edu
2	b.ergashev	$2b$12$Q4m...	Staff	b.ergashev@uni.edu

STUDENT — Student profile

PK student_id INT	FK user_id INT	first_name VARCHAR(40)	last_name VARCHAR(40)	major VARCHAR(50)
101	1	Aziza	Karimova	Computer Science
102	3	Jasur	Rahimov	Business

STAFF — Registrars, instructors and department heads

PK staff_id INT	FK user_id INT	first_name VARCHAR(40)	last_name VARCHAR(40)	position VARCHAR(30)
1	2	Bobur	Ergashev	Instructor
2	4	Dilnoza	Saidova	Registrar

Explanation These tables store who can log in and the profile of each student and staff member. Each STUDENT and STAFF row points to one USER_ACCOUNT through a foreign key (FK). PK = primary key, UQ = unique; two sample rows are shown for each table.

6.2 Database Tables 2: COURSE, SECTION and REGISTRATION

COURSE — Course catalog

PK course_id INT	UQ course_code VARCHAR(10)	course_title VARCHAR(80)	credits TINYINT	prerequisite_code VARCHAR(10)
1	CSE101	Programming Basics	3	*NULL*
2	CSE201	Database Systems	3	CSE101

SECTION — A course offered in one semester

PK section_id INT	FK course_id INT	FK instructor_id INT	day_time VARCHAR(30)	room VARCHAR(10)	capacity INT	seats_taken INT
1	2	1	Mon/Wed 09:00-10:30	A-301	40	38
2	1	1	Tue/Thu 11:00-12:30	A-105	30	30

REGISTRATION — A student's enrollment in a section

PK registration_id INT	FK student_id INT	FK section_id INT	registration_date DATE	status VARCHAR(12)
1	101	1	2026-09-01	Registered
2	102	2	2026-09-01	Waitlisted

Explanation COURSE is the catalog, SECTION is a course offered with its time, room, capacity and used seats, and REGISTRATION links a student to a section with a status (Registered, Waitlisted or Dropped). NULL means that the course has no prerequisite.

7 Consistency and Conclusion

The table links each use case to its business process, its main database tables and its screen.

Use case	BPM	Main tables	Screen
UC-01 Log In	-	USER_ACCOUNT	SCR-1
UC-02 Register for Course	BPM-2	REGISTRATION, SECTION, COURSE, STUDENT	SCR-2
UC-03 Drop Course	BPM-3	REGISTRATION, SECTION	-
UC-04 Manage Sections	-	SECTION, COURSE, STAFF	SCR-3
UC-05 View Enrollment Report	-	SECTION, REGISTRATION	-
UC-06 Manage User Accounts	-	USER_ACCOUNT, STUDENT, STAFF	-

Explanation Actor, use case, entity and table names are the same in every diagram, so the models agree with each other.

7.1 Conclusion

The project analyzed how course registration works today and modeled the proposed system with a use case diagram, three business process models, a Chen and an IE ERD, three screen layouts and six database tables. Names of actors, use cases and tables are the same in every diagram, so the models agree with each other and can be used as the base for later development.

7.2 References

Course lecture notes (Cap1: System Analysis and Design).

Dennis, A., Wixom, B. H., & Roth, R. M. (2012). Systems Analysis and Design (5th ed.). Wiley.

Elmasri, R., & Navathe, S. B. (2016). Fundamentals of Database Systems (7th ed.). Pearson.